

Simultaneous Illumination Normalization and Sparse Image Compression via Unfolded Recovery

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Background & Motivation

Cameras sample then throw away

A JPEG keeps only 5–10% of DCT coefficients, and the ISP further reshapes the data through demosaicing, tone-mapping, and white-balance. By the time bits leave the sensor, most of what was measured is gone — yet acquiring every pixel at the sensor still dominates power and shutter time on mobile and embedded platforms.

Compressed sensing reverses the order

Instead of sample-then-throw-away, take $M \ll N$ random linear measurements and reconstruct via sparsity. The CS framework is provably exact when the signal is sparse in some basis Ψ and the sensing matrix $\mathbf{A}$ is incoherent with it [3, 4] — the same principle that underpins single-pixel cameras, snapshot hyperspectral imaging, and accelerated MRI.

Forward model and Lasso recovery

With $\mathbf{x} = \Psi \mathbf{c}$ and $\|\mathbf{c}\|_0 \leq S$, the forward model is $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}$, $\mathbf{A} \in \mathbb{R}^{M \times N}$. If $\mathbf{A}$ satisfies the RIP with $\delta_{2S} < \sqrt{2} - 1$, the Lasso relaxation

$$\mathbf{c}_\lambda = \arg \min_{\mathbf{c}} \frac{1}{2} \|\mathbf{A}\Psi \mathbf{c} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{c}\|_1$$

recovers $\mathbf{c}$ to within $O(\epsilon)$ [3]. We use the 2D DCT as Ψ : natural images concentrate $\geq 90\%$ of their energy in $< 10\%$ of low-frequency coefficients.

Symbols at a glance

$\mathbf{s}$ – reflectance (sparse in DCT); $\mathbf{g}$ – illumination (smooth); $\mathbf{x} = \mathbf{s} \odot \mathbf{g}$ – RAW pixels; $\mathbf{y}$ – compressed measurements; $\mathbf{A} \in \mathbb{R}^{M \times N}$ – Gaussian sensing matrix; Ψ – 2D DCT basis; $\mathbf{c} = \Psi^\top \mathbf{s}$ – sparse coefficients; $\delta = M/N$ – measurement rate; $\rho = S/M$ – sparsity rate.

Problem

RAW sensors measure $\mathbf{x} = \mathbf{s} \odot \mathbf{g}$, the *product* of a sparse reflectance $\mathbf{s}$ and a smooth illumination field $\mathbf{g}$. The conventional **CS-then-correct** pipeline (i) reconstructs $\mathbf{x}$ from $\mathbf{y}$ by enforcing ℓ_1 on $\mathbf{x}$, then (ii) divides by an illumination estimate (typically a column mean) to recover $\mathbf{s}$.

Two errors compound. Sparsity is broken at the prior: $\mathbf{x}$ is *not* DCT-sparse because point-wise multiplication by $\mathbf{g}$ smears DCT energy across coefficients; ℓ_1 then over-shrinks high-illumination coefficients before correction ever runs. **Correction is shape-rigid:** a column-mean divisor copies the gradient but cannot restore information already shrunk away.

The sequential pipeline wastes its prior – every ℓ_1 iteration penalises structure that wasn't really there to begin with. **Naive fixes don't help:** whitening the measurements requires knowing $\mathbf{g}$ in advance, and pre-flattening assumes a parametric form that real scenes routinely violate.

Proposed Solution

Estimate reflectance and illumination **jointly** via a single ℓ_1 +smooth-prior objective in $(\mathbf{c}, \mathbf{g})$, solved by block-coordinate alternation. A $K=10$ -layer LISTA replaces the FISTA inner loop for fast, learned inference.

Why joint helps. The illumination $\mathbf{g}$ is smooth (a Tikhonov prior fits it) and the reflectance $\mathbf{s}$ is sparse in DCT (an ℓ_1 prior fits it). Estimating both at once lets each prior act on the signal it was designed for, instead of forcing one ℓ_1 on the compromised product $\mathbf{x} = \mathbf{s} \odot \mathbf{g}$.

Joint recovery objective. A single ℓ_1 +smooth-prior problem in $(\mathbf{c}, \mathbf{g})$: FISTA / LISTA on $\mathbf{c}$, closed-form ridge on $\mathbf{g}$.

Deep-unfolded inner solver. $K=10$ -layer LISTA [6] with Adam, gradient clipping, best-checkpoint restore for stability.

Scene-dependent benchmark. Phase transition, multi-image rate-distortion, LISTA-vs-classical, per-scene illumination ablation on 5 natural images.

Honest failure-mode analysis. Joint catastrophically fails on near-uniform scenes (moon, -8 dB); traced to scene-energy deficiency in the first c-step, suggesting a deployment-time energy gate.

Headline result. +1.1 to +4.7 dB on textured scenes (cameraman, astronaut); 2.3 $\times$ matched iteration reduction from five-seed LISTA, but no CPU latency gain.

Architecture

Encoder $\rightarrow$ Decoder pipeline

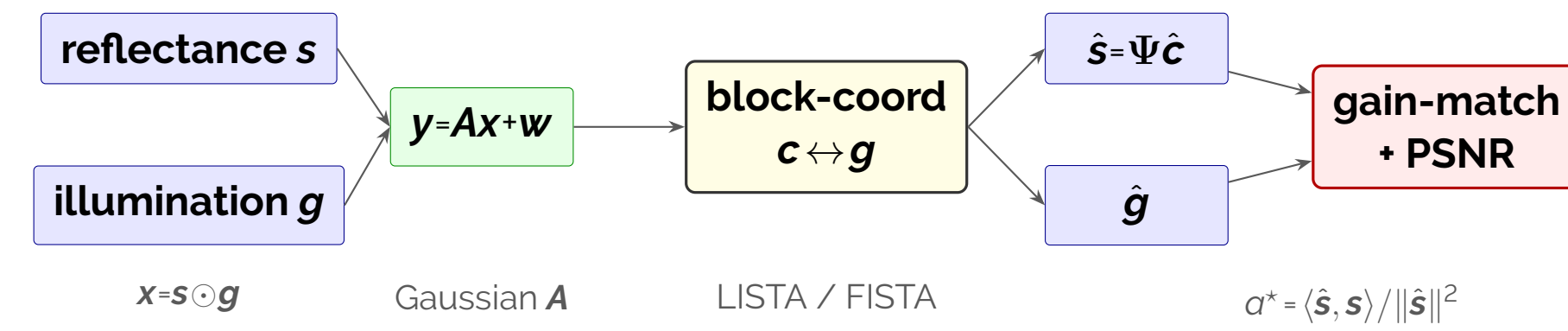


Figure 1. End-to-end joint recovery pipeline. The RAW signal $\mathbf{x} = \mathbf{s} \odot \mathbf{g}$ is measured via Gaussian $\mathbf{A}$; block-coordinate alternation recovers $(\mathbf{c}, \mathbf{g})$ simultaneously; the scalar gain ambiguity is resolved at evaluation.

Joint recovery objective

Replace the two-stage pipeline with a single objective:

$$\min_{\mathbf{c}, \mathbf{g}} \frac{1}{2} \|\mathbf{A} \text{diag}(\mathbf{g}_{\text{pix}}) \Psi \mathbf{c} - \mathbf{y}\|_2^2 + \lambda_c \|\mathbf{c}\|_1 + \lambda_g \|\nabla \mathbf{g}\|_2^2.$$

Non-convex jointly but **block-convex** in each variable: the $\mathbf{c}$ -step is a standard Lasso (FISTA / LISTA); the $\mathbf{g}$ -step is closed-form ridge.

Block-coordinate recipe

Initialise $\mathbf{g}^{(0)} = \mathbf{1}$, $\mathbf{c}^{(0)} = \mathbf{0}$. For $t=1, \dots, T_{\text{outer}}=8$: the **c-update** runs T_c FISTA (or K LISTA) steps on $\mathbf{A}_{\text{diff}}(\mathbf{g}^{(t-1)})$; the **g-update** solves $(\mathbf{B}^\top \mathbf{B} + \lambda_g \mathbf{L}) \mathbf{g} = \mathbf{B}^\top \mathbf{y}$ in closed form, clamping to $[\epsilon_g, \infty)$.

Closed-form $\mathbf{g}$ -step

With $\hat{\mathbf{s}} = \Psi \mathbf{c}$ the model is linear in $\mathbf{g}$, $\mathbf{y} = \mathbf{B}(\hat{\mathbf{s}}) \mathbf{g} + \mathbf{w}$, giving

$$\mathbf{g}^* = (\mathbf{B}^\top \mathbf{B} + \lambda_g \mathbf{L})^{-1} \mathbf{B}^\top \mathbf{y},$$

with $\mathbf{L} = \nabla^\top \nabla$ the 1D discrete Laplacian. The scalar invariance $(\mathbf{s}, \mathbf{g}) \rightarrow (a\mathbf{s}, \mathbf{g}/a)$ is resolved at evaluation by the optimal positive gain $\alpha^* = \langle \hat{\mathbf{s}}, \mathbf{s}_{\text{true}} \rangle / \|\hat{\mathbf{s}}\|_2^2$, applied identically to baselines.

Deep-unfolded LISTA inner solver

Replace the FISTA $\mathbf{c}$ -step with K learned layers [6]:

$$\mathbf{c}^{(k+1)} = \mathcal{S}_{\theta_k}(\mathbf{W}_t \mathbf{c}^{(k)} + \mathbf{W}_e \mathbf{y}), \quad k = 0, \dots, K-1.$$

Initialise $\mathbf{W}_e = \frac{1}{L} (\mathbf{A} \Psi)^\top$, $\mathbf{W}_t = \mathbf{I} - \frac{1}{L} (\mathbf{A} \Psi)^\top \mathbf{A} \Psi$, $\theta_k = \lambda/L$ to reproduce ISTA at step zero, then train end-to-end on $(\mathbf{y}, \mathbf{c})$ pairs.

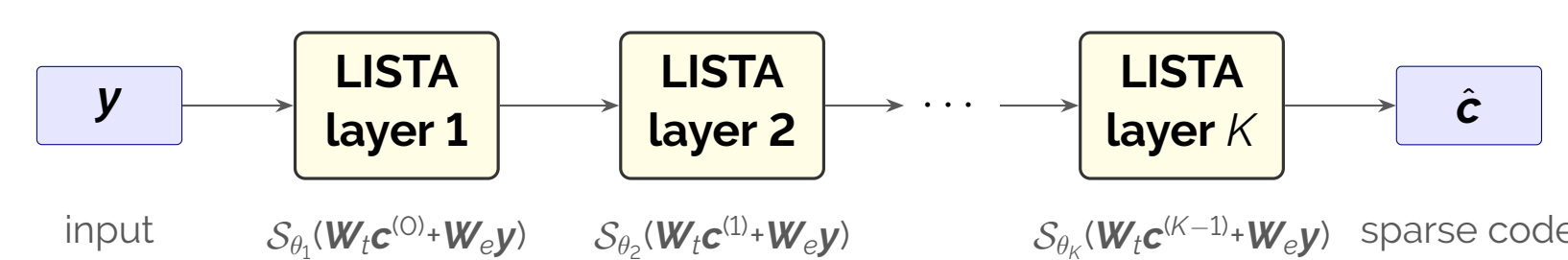


Figure 2. K -layer LISTA with tied weights $\mathbf{W}_e$, $\mathbf{W}_t$ and per-layer learnable thresholds θ_k . Each block is one ISTA-style update followed by soft-thresholding; the recurrence reproduces ISTA at initialisation.

Stability tricks

Naive Adam at 5×10^{-3} produces a clean descent to 0.12 NMSE, then blows up to $\sim 10^3$. Diagnosing, $\rho(\mathbf{W}_t) > 1$ along some eigendirection causes the recurrence to diverge. The fix combines (i) lowering $\text{lr} \rightarrow 10^{-3}$, (ii) gradient clip-norm 1.0, and (iii) restoring the best-validation checkpoint.

LISTA vs. classical at matched compute

Setup. $K=10$ tied-weight LISTA trained on 5000 synthetic $(\mathbf{y}, \mathbf{c})$ pairs with $N=200$, $M=80$, $S=10$, SNR=30 dB.

Solver	Iters/Layers	Val NMSE
ISTA	10	0.467
FISTA	10	0.337
LISTA	10	0.0435$\pm$0.0120
FISTA (matched)	23	$\approx$ 0.0435
FISTA (conv.)	500	0.005

2.3 $\times$ **iteration reduction**, but measured batch-1 CPU latency is 0.664 ms (LISTA) vs. 0.484 ms (FISTA).

Results & Benchmarks

Phase transition

Empirical Donoho–Tanner boundary on synthetic Gaussian CS, $N=200$, 20 trials/cell, NMSE $< 10^{-3}$.

$\delta = M/N$	0.30	0.50	0.70	0.90
OMP ρ^*	0.35	0.40	0.40	0.50
FISTA ρ^*	0.20	0.30	0.50	0.65

OMP wins low δ , FISTA wins high δ ; crossover near $\delta \approx 0.65$.

Rate-distortion

5-image natural-scene set, SNR=30 dB, 16×16 blocks. Mean PSNR $\pm$ std across scenes.

δ	0.20	0.40	0.50	0.70
OMP	17.4 $\pm$ 4.3	20.4 $\pm$ 3.7	21.7 $\pm$ 3.2	24.9 $\pm$ 3.0
FISTA	19.4$\pm$4.5	23.6$\pm$4.1	25.1$\pm$3.9	27.6$\pm$3.5
ADMM	17.6 $\pm$ 4.8	23.5 $\pm$ 4.4	25.0 $\pm$ 4.0	27.5 $\pm$ 3.7

FISTA and ADMM agree within 0.5 dB (same Lasso objective) and both beat OMP by 2–3 dB; wide std reflects scene difficulty.

Joint vs. Sequential: Per-Scene PSNR Gain (dB)

$8 \times$ horizontal illumination gradient, SNR=25 dB, 16×16 center patch per scene. $\Delta = \text{PSNR}_{\text{joint}} - \text{PSNR}_{\text{seq}}$.



Figure 3. Per-scene PSNR gain over the sequential baseline. Textured wins +1.1 to +4.7 dB on cameraman, astronaut; **moon collapses by ~ 8 dB** – a real, mechanism-explainable failure mode.

Limitations

Failure mode: near-uniform scenes

Moon's centre patch is mostly black sky after $8 \times$ gradient multiplication, so the first c-step has too little scene energy; noise then dominates the gain update and drives $\mathbf{g}$ to a noise-fitting solution. The sequential pipeline avoids this by *never* estimating $\mathbf{g}$ – its column-mean divisor is essentially a copy of the gradient. **When the scene cannot identify $\mathbf{g}$, refusing to estimate it is a virtue.** A $\kappa(\mathbf{B}^\top \mathbf{B})$ gate was tested and falsified. **Deployment rule:** gate on the first c-step's scene energy $\|\hat{\mathbf{s}}\|_2$ and fall back to sequential when it is too small.

Other limitations

The 1D illumination model suits horizontal gradients (2D vignettes need a richer prior); synthetic-trained LISTA needs distribution-wide noise training; coded-aperture validation and outer-loop unfolding are future work.

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